



Unit Test - Form B

Math and Measurement

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

Directions: Select the best answer for each question. Mark one choice only. Complete this test after all lessons and the cumulative review.

1. Which observation is qualitative?

- A. The crystal is colorless.
- B. The temperature is 22.4 °C.
- C. The mass is 6.250 g.
- D. The volume is 15.0 mL.

2. Which tool should be used first to measure the mass of a dry solid?

- A. Graduated cylinder
- B. Electronic balance
- C. Beaker
- D. Meter stick

3. Which record uses the correct symbol and spacing for milliliters?

- A. 6.0ml
- B. 6.0 mls
- C. 6.0 ML
- D. 6.0 mL

4. Treating 7400 micrograms as two significant figures, what is the mass in milligrams?

- A. 0.0074 mg
- B. 0.74 mg
- C. 7.4 mg
- D. 74 mg

5. What is 8.12×10^5 in standard notation?

- A. 0.000812
- B. 81,200
- C. 812,000
- D. 8,120,000



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6. How many significant figures are in 1.0200?

- A. 3
- B. 4
- C. 2
- D. 5

7. What is $45.60 / 2.5$ with correct significant figures?

- A. 18
- B. 18.2
- C. 18.24
- D. 20

8. What is $100.0 - 0.37$ with correct decimal-place precision?

- A. 99.63
- B. 99.6
- C. 100
- D. 99.630

9. What is 3.60 days in hours?

- A. 0.150 h
- B. 15.0 h
- C. 72.0 h
- D. 86.4 h

10. What is 18.0 m/s in km/h?

- A. 5.00 km/h
- B. 64.8 km/h
- C. 180 km/h
- D. 648 km/h



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11. A sample has mass 44.7 g and volume 15.0 mL. What is its density?

- A. 0.336 g/mL
- B. 2.68 g/mL
- C. 2.98 g/mL
- D. 670 g/mL

12. A solid has mass 14.9 g and density 2.50 g/cm³. What is its volume?

- A. 5.96 cm³
- B. 12.4 cm³
- C. 17.4 cm³
- D. 37.3 cm³

13. Experimental mass is 24.0 g; accepted mass is 25.0 g. What is percent error?

- A. 1.00%
- B. 4.00%
- C. 24.0%
- D. 96.0%

14. Measurements are spread out but their mean equals the accepted value. Which description fits best?

- A. Precise and accurate
- B. Precise but not accurate
- C. Neither precise nor accurate
- D. Accurate on average but not precise

15. Measurements cluster tightly above the accepted value. Which description fits best?

- A. Accurate but not precise
- B. Precise but not accurate
- C. Both accurate and precise
- D. Neither can be judged



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16. An empty balance reads +0.20 g. What type of problem is this?

- A. A systematic offset
- B. Random variation
- C. A significant-figure rule
- D. A qualitative observation

17. Why are repeated trials useful?

- A. They guarantee accuracy.
- B. They remove every bias.
- C. They reveal the spread and support a precision judgment.
- D. They make units unnecessary.

18. Why does the exact factor $1000\text{ m} = 1\text{ km}$ not limit significant figures?

- A. It is a defined relationship.
- B. It is larger than one.
- C. It contains a zero.
- D. It is written as a fraction.

19. Compared with 12.3 mL, what does 12.30 mL communicate?

- A. A larger volume
- B. A different unit
- C. A less accurate tool
- D. One additional recorded decimal place

20. Which answer is complete for a calculated volume?

- A. 5.96 cm^3
- B. 5.96
- C. volume is small
- D. about six